



- 1 a) We observe that as the temperature θ decreases, the peak of the Helmholtz free energy is much sharper, and the potential difference between the local extrema in $u = 0$ and the minima in $u \in \{-1, 1\}$ is greater at lower temperatures. We also see that the peak disappears almost completely for temperatures $\theta > \theta_c$, which theoretically marks the lowest temperature at which phase separation can occur. At $\theta < \theta_c$, the peak separates the different phases, tending to different potential wells.

- b) From the Cahn-Hilliard equation, given by

$$\partial_t u - \nabla \cdot (M \nabla (f(u) - \kappa \nabla^2 u)) = 0, \quad (1)$$

where $u = u(\vec{x}, t) \mid \vec{x} = (x, y)$, we will observe step by step what happens under the transformation $u \rightarrow -u$.

By the linearity of the derivative, we have that

$$\partial_t u \rightarrow \partial_t(-u) = -\partial_t u \quad (2)$$

$$\nabla^2 u \rightarrow \nabla^2(-u) = -\nabla^2 u, \quad (3)$$

$$(4)$$

and since f is an anti-symmetric function, we have that

$$f(u) \rightarrow f(-u) = -f(u) \quad (5)$$

$$\implies M \nabla (f(u) - \kappa \nabla^2 u) \rightarrow M \nabla (f(-u) + \kappa \nabla^2 u) = -M \nabla (f(u) - \kappa \nabla^2 u) \quad (6)$$

$$\implies \nabla \cdot (M \nabla (f(u) - \kappa \nabla^2 u)) \rightarrow -\nabla \cdot (M \nabla (f(u) - \kappa \nabla^2 u)). \quad (7)$$

Hence, we have that the equation as a whole transforms into,

$$\partial_t u - \nabla \cdot (M \nabla (f(u) - \kappa \nabla^2 u)) = 0 \quad (8)$$

$$\rightarrow -\partial_t u + \nabla \cdot (M \nabla (f(u) - \kappa \nabla^2 u)) = 0 \quad (9)$$

$$\partial_t u - \nabla \cdot (M \nabla (f(u) - \kappa \nabla^2 u)) = 0 \quad (10)$$

Thus, $-u$ also solves the Cahn-Hilliard equation and is invariant under transformation.

q.e.d.

- c) Taking the Cahn-Hilliard equation as stated above, multiplying by 1, and integrating over $\Omega = [0, L_x) \times [0, L_y)$, we get,

$$\int_{\Omega} \partial_t u \, d\vec{x} - \int_{\Omega} \nabla \cdot (M \nabla (f(u) - \kappa \nabla^2 u)) \, d\vec{x} = 0. \quad (11)$$

The first integral has an integration domain independent of the partial derivative operator inside the integral, and therefore we can move the derivative operator outside and make it a total derivative,

$$\int_{\Omega} \partial_t u \, d\vec{x} = \frac{d}{dt} \int_{\Omega} u \, d\vec{x}. \quad (12)$$

Continuing, we want to use the divergence theorem, which is generally stated as

$$\int_{\Omega} \nabla \cdot \mathbf{F} \, dV = \oint_{\partial\Omega} \hat{n} \cdot \mathbf{F} \, dS, \quad (13)$$

with $\hat{n}$ being the outward-pointing unit normal to the boundary $\partial\Omega$, and dS representing the surface element of the boundary. In our case, we can define

$$F(x, y) = M \nabla (f(u) - \kappa \nabla^2 u) \quad (14)$$

and by the periodicity of our domain, we have that $F(0, y) = F(L_x, y)$ and $F(x, 0) = F(x, L_y)$. Taking a look at the boundary and its direction, plotted here,

we clearly see that the integral over the left and right boundary segments will cancel out, and likewise with the top and bottom of the boundary. Thus

$$\oint_{\partial\Omega} \hat{n} \cdot \mathbf{F} \, dS = 0 \quad (15)$$

and we are left with

$$\frac{d}{dt} \int_{\Omega} u \, d\vec{x} = 0. \quad (16)$$

The system is therefore mass-conservative, since we showed that the boundary integral, which could be considered the mass-flux through our domain is, in total, zero.

q.e.d.

- 2** a) By Fourier-transformation of the biharmonic equation with a linear term we get

$$\Delta^2 u + cu = f \quad (17)$$

$$\mathcal{F}[\Delta^2 u + cu] = \hat{f} \quad (18)$$

$$\mathcal{F}[\Delta^2 u] + c\hat{u} = \hat{f} \quad (19)$$

$$(i\tilde{k})^4 \hat{u} + c\hat{u} = \hat{f} \quad (20)$$

$$(|\tilde{k}|^4 + c)\hat{u} = \hat{f}. \quad (21)$$

So, our operator transforms as $\{\Delta^2 + c\} \rightarrow \{|\tilde{k}|^4 + c\}$ under the Fourier transform, which will be implemented in the code under.

- b) To analyze our solvers credibility, we want to compute the Experimental Order of Convergence (EOC), defined as

$$\text{EOC}(i) = \frac{\log(e(\tau_i)) - \log(e(\tau_{i-1}))}{\log(\tau_i) - \log(\tau_{i-1})} = \frac{\log\left(\frac{e(\tau_i)}{e(\tau_{i-1})}\right)}{\log\left(\frac{\tau_i}{\tau_{i-1}}\right)} \quad (22)$$

with $e(\tau_i)$ being the maximum error at the step size τ_i . This is used to estimate the order of convergence for our methods, and also later, verify that our implemented methods corresponds to the correct theoretical methods. For this task, the step size τ_i refers to the distances dx and dy , but for later tasks it refers to the time steps dt .

- c) In our EOC analysis we can make some key observations straight away. For the first manufactured solution, $u(x, y) = \sin(8(x - 1)) \cos(4y)$, we observe a rather large error for the first couple of N_x values. A sudden change occurs from $N_x = 15$ to $N_x = 16$, with a drastic drop in error, and the EOC for 16 is consequently quite large. For $N_x > 16$ we see some random fluctuations in the error, due to floating point representation errors.

The large error for small N_x is theoretically explained by a phenomenon called aliasing. Aliasing occurs when the sampling of a function is not fine enough to capture the highest frequency fluctuations of the function. When the sampling frequency is too low, the high frequency components will be indistinguishable from lower frequency components, and the coefficients in Fourier space will occur at the same frequency coordinate, and their coefficients will add together, creating data inconsistent with the analytical expression the data was sampled from. By the Nyquist sampling theorem, you avoid aliasing issues by having a sampling rate $N \geq 2p$, where p is the highest frequency present in the analytical expression.

Employing this to $u(x, y) = \sin(8(x - 1)) \cos(4y)$, we see that the frequencies are 4 in the y-direction and 8 in the x-direction, and the Nyquist sampling rate becomes $N \geq 2 \cdot 8 = 16$, which is exactly what is observed in our EOC analysis. Moving to the first two sets of plots, both for the manufactured solution $u(x, y) = \sin(8(x - 1)) \cos(4y)$, we clearly observe the periodic accumulation of error due to aliasing for $N_x = 15$. This disappears completely in the error plot for $N_x = 16$, as previously discussed.

Now, considering the second manufactured solution, $u(x, y) = \exp(\sin^2(x) + \cos(2y))$, we calculate the Nyquist sampling rate to be $N \geq 2 \cdot 2 = 4$, so all our used values of N_x will avoid aliasing issues. We do however observe a decline in error, and what by first glance seems to be a linear increase in EOC. From its definition, a linear increase in the EOC should correspond to an exponential decrease in the error. We have therefore done a simple curve fit to illustrate this.

INSERT CURVE FIT

As we see, the error follows a somewhat exponential decay as N_x increases, which is quite characteristic for spectral and pseudo-spectral methods.

Moving to the surface plots for this manufactured solution, we clearly see that the error follows the same pattern as the function itself, which is because the timestep for steeper changes will not be able to capture the change as well as for the less steep values of the function.

- 3 a) We are given the equation

$$U_{n+1} = U_n + \tau (\theta F(t_{n+1}, U_{n+1}) + (1 - \theta) F(t_n, U_n)), \quad \text{for } n = 0, 1, \dots, N - 1 \quad (23)$$

where

$$t_{n+1} = t_n + \tau \quad (24)$$

and have that

$$k_1 = F(t_n, U_n) \quad (25)$$

$$k_2 = F(t_{n+1}, U_{n+1}). \quad (26)$$

For k_2 , we insert the expression for U_{n+1}

$$k_2 = F(t_n + \tau, U_n + \tau(\theta k_2 + (1 - \theta)k_1)) \quad (27)$$

This gives us a butcher table as follows

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & 1 - \theta & \theta \\ \hline & 1 - \theta & \theta \end{array}$$

From the butcher table, we can easily check the consistency order of the θ -method. The $p = 1$ condition, which says

$$\sum_{i=1}^2 b_i = (1 - \theta) + \theta = 1 \quad \forall \theta, \quad (28)$$

is satisfied. Now taking a look at $p = 2$, which states

$$\sum_{i=1}^2 b_i c_i = (1 - \theta) \cdot 0 + \theta \cdot 1 = \theta \neq \frac{1}{2} \quad \forall \theta, \quad (29)$$

we see that the condition is not generally satisfied. It does however hold for $\theta = \frac{1}{2}$. So assuming $\theta = \frac{1}{2}$ going forward, we compute the condition for $p = 3$,

$$\sum_{i=1}^2 b_i c_i^2 = \frac{1}{2} \cdot 0^2 + \frac{1}{2} \cdot 1^2 = \frac{1}{2} \neq \frac{1}{3}. \quad (30)$$

So the condition for consistency order $p = 3$ does not hold. Therefore, we conclude that $p = 1 \quad \forall \theta \neq \frac{1}{2}$ and $p = 2 \mid \theta = \frac{1}{2}$

For $\theta = \frac{1}{2}$, the method reduces to the Crank-Nicolson method, which is based on the implicit trapezoid rule. For $\theta = 0$, we get the explicit Euler method, and for $\theta = 1$ we are left with the implicit Euler method.

- b)** We use the butcher table for our θ -method, and the ODE $\partial_t U_n = \lambda U_n = F(t_n, U_n)$ to get,

$$U_{n+1} = U_n + \tau(\theta \lambda U_{n+1} + (1 - \theta) \lambda U_n) \quad (31)$$

$$= (1 + \tau(1 - \theta) \lambda U_n + \tau \theta \lambda U_{n+1}) \quad (32)$$

$$(1 - \theta z) U_{n+1} = (1 + (1 - \theta) z) U_n \quad (33)$$

$$U_{n+1} = \left(\frac{1 + (1 - \theta) z}{1 - \theta z} \right) U_n = \left(\frac{1 + (1 - \theta) z}{1 - \theta z} \right)^n U_0 \quad (34)$$

So our stability function becomes

$$r_\theta(z) = \frac{1 + (1 - \theta)z}{1 - \theta z} \quad (35)$$

with $z = \tau\lambda$. Below, we have included some heatmaps of the magnitude of the stability function in the complex plane.

From the heatmaps, we clearly see circular domains for low values of θ , where the boundary $|r_\theta(z)| = 1$ has been highlighted in purple, and the region of convergence is marked in blue. $\theta = 0$ gives a circle with center in $z = -1$ and radius of 1, this is in accordance with our pervious statement that $\theta = 0$ corresponds to the explicit Euler method. As θ increases, so does the radius of the circle, while the intersection point with the boundary and the origin remains fixed. When θ approaches 0.5 the radius seems to go towards infinity, and when $\theta > 0.5$ the boundary seems to bend towards the positive real axis, and we get a circular boundary where we have convergence outside of the circle. The radius keeps decreasing, and at $\theta = 1$ we have a circular boundary with radius 1 and center in $z = 1$, in accordance with our previous statement that $\theta = 1$ corresponds to the implicit Euler method.

When it comes to A-stability, we know that a method is A-stable if $\forall z \mid |z| \leq 1, \operatorname{Re}\{z\} < 0$. From the plots, the method seems to be A-stable $\forall \theta \geq 0.5$.

For general θ we expect that the boundary should be described by a circular equation, with diverging radius as $\theta \rightarrow 0.5_{\pm}$. Since the radius should be 0 at $z = 1$ it is plausible that it may depend on the argument of z , and have a denominator which is 0 at $\theta = 0.5$. The simplest denominator one could think of that satisfy this would be a linear one, on the form $2\theta - 1$.

- c) To find an expression for $\partial S_\theta = \{z \in \mathbb{C} \mid |r_\theta(z)| = 1\}$ we use the stability function and multiply by the complex conjugate to find how the radius on the boundary depends on the argument of z and θ .

$$|1 + (1 - \theta)z| = |1 - \theta z| \quad (36)$$

$$(+(1 - \theta)z) (+(1 - \theta)\bar{z}) = (1 - \theta z)(1 - \theta\bar{z}) \quad (37)$$

$$1 + (1 - \theta)(\bar{z} + z) + (1 - \theta)^2 z\bar{z} = -\theta(z + \bar{z}) + \theta^2 z\bar{z} \quad (38)$$

$$(1 - \theta)(z + \bar{z}) + (1 - \theta)^2 |z|^2 = -\theta(z + \bar{z}) + \theta^2 |z|^2 \quad (39)$$

$$(\bar{z} + z) + |z|^2 - 2\theta |z|^2 + \theta^2 |z|^2 = \theta^2 |z|^2 \quad (40)$$

$$2\operatorname{Re}\{z\} = (2\theta - 1)|z|^2 \quad (41)$$

$$|z|^2 = \frac{2\operatorname{Re}\{z\}}{2\theta - 1}. \quad (42)$$

Rewriting in polar coordinates, $z = |z|e^{i\phi}$ and solving for $|z|$ we get,

$$|z|^2 = \frac{2|z| \cos \phi}{2\theta - 1} \quad (43)$$

$$|z| = \frac{2 \cos \phi}{2\theta - 1}. \quad (44)$$

This expression for the boundary, parameterized by the argument of z , corresponds perfectly with the plotted boundaries in the heatmaps perviously discussed, as shown in the following plot.

To see how the center depends on θ , we find the real part by evaluating $\frac{|z|}{2}|_{\phi=0}$,

$$R_{center} = \frac{\cos 0}{2\theta - 1} = \frac{1}{2\theta - 1} \quad (45)$$

- d) After Fourier transforming the equation, we use the θ -method to calculate each new value for U_{n+1} . We have

$$\hat{U}_{n+1} = \hat{U}_n + \tau \left(\theta F(t_{n+1}, \hat{U}_{n+1}) + (1 - \theta) F(t_n, \hat{U}_n) \right) \quad (46)$$

where,

$$F(t_{n+1}, \hat{U}_{n+1}) = \partial_t U_{n+1} = \hat{g}(X, Y, t_{n+1}) - \kappa |\tilde{k}|^4 \hat{U}_{n+1} \quad (47)$$

$$F(t_n, \hat{U}_n) = \partial_t U_n = \hat{g}(X, Y, t_n) - \kappa |\tilde{k}|^4 \hat{U}_n \quad (48)$$

Putting this into the equation gives us

$$\hat{U}_{n+1} = \hat{U}_n + \tau \left(\theta \left(\hat{g}(X, Y, t_{n+1}) - \kappa |\tilde{k}|^4 \hat{U}_{n+1} \right) + (1 - \theta) F(t_n, \hat{U}_n) \right) \quad (49)$$

$$\Rightarrow \hat{U}_{n+1} + \tau \theta \kappa |\tilde{k}|^4 \hat{U}_{n+1} = \hat{U}_n + \tau \left(\theta \hat{g}(X, Y, t_{n+1}) + (1 - \theta) F(t_n, \hat{U}_n) \right) \quad (50)$$

$$\Rightarrow \hat{U}_{n+1} = \frac{\hat{U}_n + \tau \left(\theta \hat{g}(X, Y, t_{n+1}) + (1 - \theta) F(t_n, \hat{U}_n) \right)}{1 + \tau \theta \kappa |\tilde{k}|^4} \quad (51)$$

$$\Rightarrow \hat{U}_{n+1} = \frac{\hat{U}_n + \tau \left(\theta \hat{g}(X, Y, t_{n+1}) + (1 - \theta) (\hat{g}(X, Y, t_n) - \kappa |\tilde{k}|^4 \hat{U}_n) \right)}{1 + \tau \theta \kappa |\tilde{k}|^4} \quad (52)$$

where $\kappa |\tilde{k}|^4$ is the K_op operator. We use this to calculate the value for $\hat{U}$ for each timestep, using the value of $\hat{U}$ for the previous timestep and yield each value to create a generator.

- e) In our EOC study, we mostly observe the expected results discussed in the theoretical tasks. For $\theta = 1.0$ we get an EOC approximately 1, which is consistent with the convergence order of the implicit Euler method. For $\theta = 0.5$ we observe an EOC around 2, as expected for the Crank-Nicholson method. However, for $\theta = 0$ our results is not in line with our previous prediction of the explicit Euler method having convergence order of 1. Some further investigation is therefore required.

In our theoretical part, we derived the expression

$$r_\theta(z) = \frac{1 + (1 - \theta)z}{1 - \theta z} \quad (53)$$

for the stability function of the θ -method, where the linear eigenvalue equation $\partial_t U = \lambda U$ were used. Employing this to the transient biharmonic equation in Fourier space we obtain,

$$\partial_t \hat{U} + \kappa |\tilde{k}|^4 \hat{U} = 0 \quad (54)$$

$$\lambda \hat{U} = -\kappa |\tilde{k}|^4 \hat{U} \quad (55)$$

$$\lambda = -\kappa |\tilde{k}|^4 \quad (56)$$

which yields the stability function,

$$r_\theta = \frac{1 - (1 - \theta)\kappa|\tilde{k}|^4\tau}{1 + \theta\kappa|\tilde{k}|^4\tau}. \quad (57)$$

The Courant-Friedrichs-Lewy (CFL) condition is in general a condition for PDEs that relates the maximum timestep allowed in order to obtain convergence given a spatial step-size in our simulation grid. This is found by employing the convergence criteria, $|r_\theta| \leq 1$, to the stability function for the θ -method of the transient biharmonic equation.

We mainly want to investigate the case of $\theta = 0$, to build some understanding around why our previous error- and EOC calculations yield such terrible results for this simulation. Note that we are working in Fourier space while solving the equation, so our CFL condition will relate the maximum frequency in Fourier space to the time-step, rather than the spacial step-size. Doing this, we see that

$$|r_0| = |1 - \kappa|\tilde{k}|_{max}^4\tau| \leq 1 \quad (58)$$

$$-1 \leq 1 - \kappa|\tilde{k}|_{max}^4\tau \leq 1 \quad (59)$$

$$1 - \kappa|\tilde{k}|_{max}^4\tau \geq -1 \quad (60)$$

$$\kappa|\tilde{k}|_{max}^4\tau \leq 2 \quad (61)$$

$$\implies \tau \leq \frac{2}{\kappa|\tilde{k}|_{max}^4} \quad (62)$$

$$\tau_{CFL} = \frac{2}{\kappa|\tilde{k}|_{max}^4}. \quad (63)$$

And thus, by the relation $N = \frac{T}{\tau}$, we get

$$N_{CFL} = \frac{T}{\tau_{CFL}} = \frac{T\kappa|\tilde{k}|_{max}^4}{2} \quad (64)$$

To avoid lots of unnecessary prints in our submitted code, we chose to omit the lines of code calculating N_{CFL} , but in a test run we implemented the following line in the transient biharmonic solver function, $N_cfl = T*np.real(np.max(K_op))/2$, which returned that $N_{CFL} = 20000$ for $\theta = 0$. Since the time-steps used in our EOC analysis earlier where in the 10 to 640 range, this explains why we didn't see any error-convergence for $\theta = 0$. The time-steps simply weren't small enough.

INSERT CODE FOR NCFL

As we see, when we reach $N_t = N_{CFL}$ our error dramatically decrease, and we obtain the expected convergence order of 1 for the explicit Euler method.

AFTER NEW EOC 3.3 Also for this manufactured solution, we observe the expected convergence orders.

f)

- 4 a) Using the given IMEX scheme for the Cahn-Hilliard equation with constant mobility $M = 1$ from the theoretical introduction to this task, but with the

slight alteration of adding an implicit evaluation of g at the right hand side, i.e.,

$$\frac{u_{n+1} - u_n}{\tau} + \kappa \Delta^2 u_{n+1} - a \Delta u_{n+1} = \Delta \left((u_n)^3 - (1 + a)u_n \right) + g_{n+1} \quad (65)$$

rewriting this equation to the form $U_{n+1} + \tau \mathbf{L} U_{n+1} = U_n + \tau \{ \mathbf{N}(U_n) + g_{n+1} \}$, we get

$$u_{n+1} + \tau \{ \kappa \Delta^2 - a \Delta \} u_{n+1} = u_n + \tau \{ \Delta \left((u_n)^3 - (1 + a)u_n \right) + g_{n+1} \}. \quad (66)$$

Here, our non-linear term and linear operator becomes

$$\mathbf{N}(u_n) = \Delta \left((u_n)^3 - (1 + a)u_n \right) \quad (67)$$

$$\mathbf{L} = \{ \kappa \Delta^2 - a \Delta \} \quad (68)$$

and our IMEX scheme can be neatly rewritten as

$$u_{n+1} + \tau \mathbf{L} u_{n+1} = u_n + \tau \{ \mathbf{N}(u_n) + g_{n+1} \} \quad (69)$$

Transforming this equation to Fourier-space, we obtain

$$(1 + \tau \hat{\mathbf{L}}) \hat{u}_{n+1} = \hat{u}_n + \tau \{ \widehat{\mathbf{N}(u_n)} + \hat{g}_{n+1} \} \quad (70)$$

$$\hat{u}_{n+1} = \frac{1}{(1 + \tau \hat{\mathbf{L}})} \left(\hat{u}_n + \tau \{ \widehat{\mathbf{N}(u_n)} + \hat{g}_{n+1} \} \right) \quad (71)$$

where the linear operator in Fourier space becomes

$$\hat{\mathbf{L}} = \{ \kappa |\tilde{k}|^4 + a |\tilde{k}|^2 \} \quad (72)$$

So to solve for the next timestep, we need to calculate the non-linear term in Fourier space. Since the product of functions becomes a convolution in Fourier space, we will use the inverse Fourier transform to calculate the non-linear term in cartesian space, and transform the solution of this calculation back to Fourier space in order to find the solution at the next timestep. The transformation scheme for calculating the non-linear term goes as follows,

$$\hat{u}_n \rightarrow u_n \rightarrow f(u_n) \rightarrow \widehat{f(u_n)} \rightarrow \widehat{\mathbf{N}(u_n)} \quad (73)$$

where going from $\hat{f}$ to $\hat{\mathbf{N}}$ we apply the Laplace operator in Fourier space.

- b)** In our convergence study, we observe a general order of convergence of approximately 1 for both values of κ , which confirms our method is of first order in time.

Furthermore, the magnitude of the error is larger for $\kappa = 1$ then $\kappa = 0.01$. This is because larger κ makes the time dependent decrease more rapid, and you thus need a finer grid to capture the rapid change, resulting in a larger error for the same N_t and a larger κ

- 5** **a)** We consider the equation

$$U_t = -\mathbf{L}U + \mathbf{N}(U), \quad U(t_0) = U_0. \quad (74)$$

with the numerical scheme

$$U^{(1)} = U^n + \tau \left(\mathbf{L}U^{(1)} + \mathbf{N}(U^n) \right), \quad (75)$$

$$U^{(2)} = \alpha_{10}U^n + \alpha_{11}U^{(1)} + \beta_1\tau \left(\mathbf{L}U^{(2)} + \mathbf{N}(U^{(1)}) \right), \quad (76)$$

$$U^{n+1} = \alpha_{20}U^n + \alpha_{21}U^{(1)} + \alpha_{22}U^{(2)} + \beta_2\tau \left(\mathbf{L}U^{n+1} + \mathbf{N}(U^{(2)}) \right) \quad (77)$$

the Taylor expansion of the exact solution is,

$$U(t_0 + \tau) = U(t_0) + U'(t_0)\tau + \frac{1}{2}U''(t_0)\tau^2 + \mathcal{O}(\tau^3), \quad (78)$$

where

$$U'(t_0) = -\mathbf{L}U_0 + \mathbf{N}(U_0), \quad U''(t_0) = U'(t_0) (\mathbf{N}'(U_0) - \mathbf{L}). \quad (79)$$

The local truncation error is

$$\eta = U(t_0 + \tau) - U^{n+1}. \quad (80)$$

Inserting the Taylor expansion and the full form of the numerical update to get:

$$\eta = \left[U_0 + \tau U'(t_0) + \frac{\tau^2}{2}U''(t_0) + \mathcal{O}(\tau^3) \right] \quad (81)$$

$$- \left[\alpha_{20}U_0 + \alpha_{21}U^{(1)} + \alpha_{22}U^{(2)} + \beta_2\tau(\mathbf{L}U^{n+1} + \mathcal{N}(U^{(2)})) \right]. \quad (82)$$

The method is consistent of order $p = 2$ if and only if

$$\eta \leq D\tau^{p+1} = D\tau^3 \quad \text{for some } D > 0, \quad (83)$$

which means that η have to be of order 3 or higher and the coefficients must therefore be determined so that the the other terms are canceled out. $U^{(1)}$ and $U^{(2)}$ can be used in the expression of η as well as the expressions of $U'(t_0)$ and $U''(t_0)$. Furthermore, we get

$$\begin{aligned} \eta = & \left[U_0 + \tau (-\mathbf{L}U_0 + \mathbf{N}(U_0)) + \frac{\tau^2}{2} (U'(t_0)(\mathbf{N}'(U_0) - \mathbf{L})) + \mathcal{O}(\tau^3) \right] \\ & - \left(\alpha_{20}U^0 + \alpha_{21} \left(U^0 + \tau \left(\mathbf{L}U^{(1)} + \mathbf{N}(U^0) \right) \right) \right. \\ & \left. - \alpha_{22} \left(\alpha_{10}U^0 + \alpha_{11}U^{(1)} + \beta_1\tau \left(\mathbf{L}U^{(2)} + \mathbf{N}(U^{(1)}) \right) \right) - \left(\beta_2\tau(\mathbf{L}U^{n+1} + \mathcal{N}(U^{(2)})) \right) \right) \end{aligned}$$

and if you keep going like that, which will not be prioritized this time, you should end up with terms where the order conditions cancel out all terms with τ of order 2 and lower which shows that the scheme has consistency order $p = 2$.

- b) This numerical scheme is happening in the Fourier space where expressions for $\hat{U}^{(1)}$, $\hat{U}^{(2)}$, and $\hat{U}^{n+1}$ are retrieved from the given equations. The operators $\hat{\mathbf{L}}$ and $\hat{\mathbf{N}}(U)$ are almost the same as in task 4. However, because $U_t = \mathbf{L}U + \mathbf{N}(U)$, the sign in the $\hat{\mathbf{L}}$ operator differs from the previous task. We get

$$U_t = \mathbf{L}U + \mathbf{N}(U) \quad (84)$$

$$\Rightarrow \frac{U_{n+1} - U_n}{\tau} = \mathbf{L}U_{n+1} + \mathbf{N}(U_n) \quad (85)$$

$$\Rightarrow U^{n+1} + \tau(-\mathbf{L})U^{n+1} = U^n + \tau\mathbf{N}(U^n). \quad (86)$$

which is the same equation used to find $\mathbf{L}$ in the Cahn-Hilliard equation, but this time with a negative sign in front of $\mathbf{L}$. Therefore, we obtain

$$\hat{\mathbf{L}} = -\{\kappa\Delta^2 - a\Delta\} \quad (87)$$

$$\hat{\mathbf{N}}(\hat{U}_n) = \Delta \left((\hat{U}_n)^3 - (1+a)\hat{U}_n \right) \quad (88)$$

where $\hat{\mathbf{N}}(\hat{U}_n)$ is the non-linear term in Fourier space. In the implementation, we inversely transform $\hat{U}$ to the cartesian space and then return $-\kappa|\tilde{k}|^2\mathcal{F}[f(U)]$ where $f(u) = U^3 - (1+a)U$ and $-\kappa|\tilde{k}|^2$ is the Laplace operator in Fourier space. This is done to avoid the convolution from the transformation of U^3 just as in the previous task. However, the operator is used in each of the steps in the numerical scheme, with different input values, and is therefore implemented as a function of its own. Further, we have

$$\hat{\mathbf{G}}^{n+1/2} = \mathcal{F}[g(X, Y, t_0 + \frac{\tau}{2})] \quad (89)$$

$$(90)$$

$$\hat{U}^{(1)} = \hat{U}^n + \tau \left(\mathbf{L}\hat{U}^{(1)} + \mathbf{N}(\hat{U}^n) + \hat{\mathbf{G}}^{n+1/2} \right) \quad (91)$$

$$\Rightarrow \hat{U}^{(1)} = \frac{\hat{U}^n + \tau \left(\mathbf{N}(\hat{U}^n) + \hat{\mathbf{G}}^{n+1/2} \right)}{1 - \tau\mathbf{L}} \quad (92)$$

$$\hat{U}^{(2)} = \alpha_{10}\hat{U}^n + \alpha_{11}\hat{U}^{(1)} + \beta_1\tau \left(\mathbf{L}\hat{U}^{(2)} + \mathbf{N}(\hat{U}^{(1)}) + \hat{\mathbf{G}}^{n+1/2} \right) \quad (93)$$

$$\Rightarrow \hat{U}^{(2)} = \frac{\alpha_{10}\hat{U}^n + \alpha_{11}\hat{U}^{(1)} + \beta_1\tau \left(\mathbf{N}(\hat{U}^{(1)}) + \hat{\mathbf{G}}^{n+1/2} \right)}{1 - \beta_1\mathbf{L}\tau} \quad (94)$$

$$\hat{U}^{n+1} = \alpha_{20}\hat{U}^n + \alpha_{21}\hat{U}^{(1)} + \alpha_{22}\hat{U}^{(2)} + \beta_2\tau \left(\mathbf{L}\hat{U}^{n+1} + \mathbf{N}(\hat{U}^{(2)}) \right) \quad (95)$$

$$\Rightarrow \hat{U}^{n+1} = \frac{\alpha_{20}\hat{U}^n + \alpha_{21}\hat{U}^{(1)} + \alpha_{22}\hat{U}^{(2)} + \beta_2\tau \left(\mathbf{L}\hat{U}^{n+1} + \mathbf{N}(\hat{U}^{(2)}) \right)}{1 - \beta_2\tau\mathbf{L}} \quad (96)$$

which is the numerical scheme used in the `IMEX_song_solver` function.

- c) We notice that the `IMEX_song_solver` give an error with dimension between 10^{-5} and 10^{-7} except one error for the last set of coefficients where $N_t = 100$ and the error is $1.154681 \cdot 10^{-4}$. For comparison, the `cahn_hilliard_backward_euler`

has errors with dimension between 10^{-2} and 10^{-4} . The EOC study however, has better numbers for the `cahn_hilliard_backward_euler`, but this is not really relevant when comparing the two different methods, as the EOC study is based on how the results differs for different values of N_t for one method.

The coefficients that gave the least error is

$$\alpha_{10} = 2, \alpha_{11} = -1, \alpha_{20} = \frac{1}{2}, \alpha_{21} = 0, \alpha_{22} = \frac{1}{2}, \beta_1 = 1, \beta_2 = 1 \quad (97)$$

and is therefore our favorite and will be used later on in the next task.

- 6 a) To calculate the total mass $\int_{\Omega} u \, dx$, we simply calculate the sum over the whole U mesh grid for each timestep.

From the exercise sheet we know that the interface energy is defined as

$$\mathcal{E}_{\text{int}}(u) = \int_{\Omega} \frac{\kappa}{2} |\nabla u|^2 \, dx. \quad (98)$$

To calculate the square of the gradient of u , we calculate the gradient's x and y components separately in the Fourier space, by multiplying the respective K meshgrid in each spatial direction. Then we sum the squares of the gradient components to find the modulo of the gradient squared, and sum over the whole resulting meshgrid and multiply by $\frac{\kappa}{2}$.

To calculate the mixing energy,

$$\mathcal{E}_{\text{mix}}(u) = \int_{\Omega} F(u) \, dx, \quad (99)$$

we use the function $F(u) = \frac{1}{4}(u^2 - 1)^2$, and take the sum over the whole meshgrid.

- b) The total mass remains numerically constant throughout the entire period. This is just as expected due to conservation of mass as shown in task 1.3. Any slight fluctuations observed are most likely due to the inherent limitations of floating-point precision, which can result in small inaccuracies in the representation of values.

The two different solvers are quite similar. However, there are some prominent differences. Firstly, there is an irregularity in the first plot with the backward Euler method and the `IMEX_ssong_solver` gives an overall smoother curvature. Some animations were created to look for any obvious differences, and the end result looks quite different, so the choice of solver definitely has an effect. Also, with bigger time steps, the `IMEX_ssong_solver` has a better representation of the spinodal decomposition whereas the backward Euler method is not able to represent this for $dt = 10^{-3}$ in the shorter time interval $T = 0.1$. We did tests for smaller time steps dt for both solvers and saw that the spinodal decomposition times for the `IMEX_ssong_solver` stabilized faster (around $dt = 10^{-6}$) than the backward Euler method. This is another reason for why we decided the `IMEX_ssong_solver` was better at simulating the given scenario. Further, for the backward Euler method the interface energy starts around 10^3 and rapidly drops for the first milliseconds before spiking as initially expected. This is

not expected behaviour for the interface energy, which will be explained further on, and therefore indicates that there are some weaknesses with the numerical method. Therefore, the `IMEX_song_solver` is used in further evaluations of the different initial conditions.

As expected, the different energies decrease over time as the system seeks equilibrium. As described in the task, the mixing energy drives the system from a homogeneous state towards a separated state, while the interface energy works to minimize the surface area. Together, these two effects create complex patterns during the separation process. The total energy is governed by the dynamics of the system, aiming to minimize the system's free energy. However, at the beginning, the interface energy should experience a spike due to spinodal decomposition, where the mixture transitions from a homogeneous state and begins to separate. This is not what is observed for the backwards Euler solver and is a big reason for not choosing this solver for further analysis. Nevertheless, this behavior is reflected in the other plots, where we observe the initial spike in energy followed by a decrease over time for $T = 4$. For $T = 0.01$, the spinodal decomposition is visible in more detail, demonstrating the early-stage separation more clearly.

There are two different initial conditions for the energy, where $U_2 = -0.45 + 0.05rand(x, y)$ gives a more jagged form and $U_1 = 0.05rand(x, y)$ has a smooth curvature. This behavior, with small, sudden jumps, instead of a completely smooth settling, could be explained from the negative bias. The negative bias will result in stronger local instabilities and steeper energy gradients, which may result in the sudden downwards jumps in total energy observed in the plot. Moreover, the animations show some differences. As one of the initial conditions has a negative bias, the animation for $U_0 = U_2$ starts with a more prominent orange color, indicating a more negative relationship between the two substances compared to $U_0 = U_1$, where the color is closer to white. This is confirmed by the mass calculations, which show around 16 for $U_0 = U_1$ and around -29000 for $U_0 = U_2$, demonstrating a large skewed distribution of the substances.

When looking at the Ostwald ripening and spinodal decomposition, the difference in the slope is more prominent for the smaller time steps. Furthermore, the spinodal decomposition doesn't last as long when looking at the animations and plots made with smaller time steps, and T_{sd} is therefore marginally larger. For example, one of the plots for $U_0 = U_2$ and $T = 0.01$ did not capture the whole spinodal decomposition, because of the longer duration for bigger timestep. Therefore, the smaller time steps appears to give a more exact solution, which sounds reasonable.

Therefore, we look at the plots with the smaller time steps to determine where the spinodal decomposition and Ostwald ripening is happening. For $U_0 = U_1$, $T_{sd} = 0.0009$ approximately, and the Ostwald ripening is happening after this. For $U_0 = U_2$, T_{sd} is around 0.0022 and the Ostwald ripening, again, occurs after the spinodal decomposition.